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Exterior Lighting-VF.CONCEPT

SFRS for Zefiro-Europe

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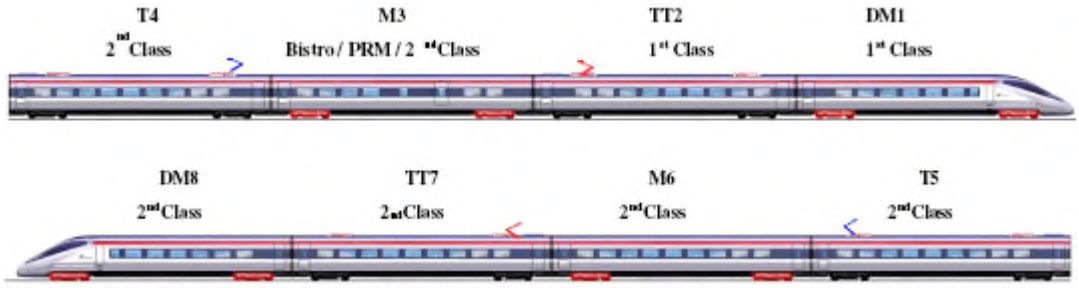
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1 Introduction

This document is a function specification for the control of the *Exterior Lighting* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

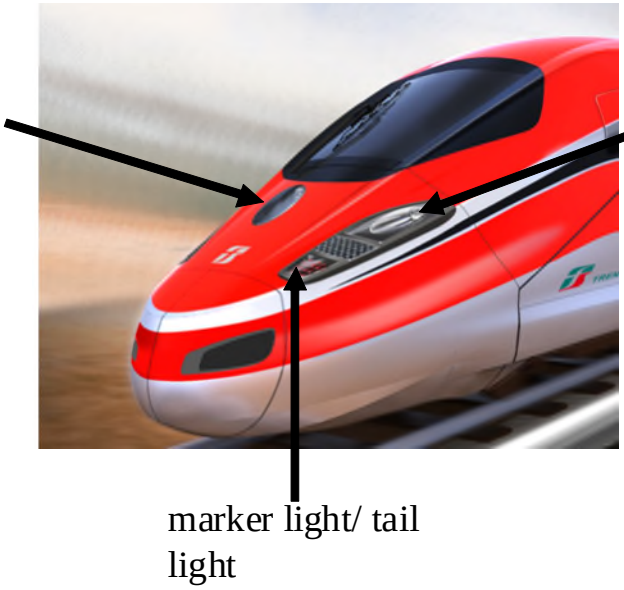
1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
HV	High voltage

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2 System Overview



Definition of the used names for the signal light system - the shown shape of the light units is not to be used for any questions of design.

Requested intensity of the light units is according to the TSI:

	Dimmed head light	Full-beam head light
Luminous intensity (cd) along centre line	12 000-16 000	> 40 000

	Dimmed marker light	Full marker light
Luminous intensity (cd) along centre line	Minimum 100	300-700

	Dimmed centre marker light	Full centre marker light
Luminous intensity (cd) along centre line	Minimum 50	150-350

	Tail light	
Luminous intensity (cd) along centre line	15-40	

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2.1 System Configuration

The exterior lighting system has 3 tasks.

1. The marking of front and rear end of the train and hereby signalling the driving direction during normal service.
2. To improve visibility for the driver
3. To mark the train when in standstill and in preparation for service

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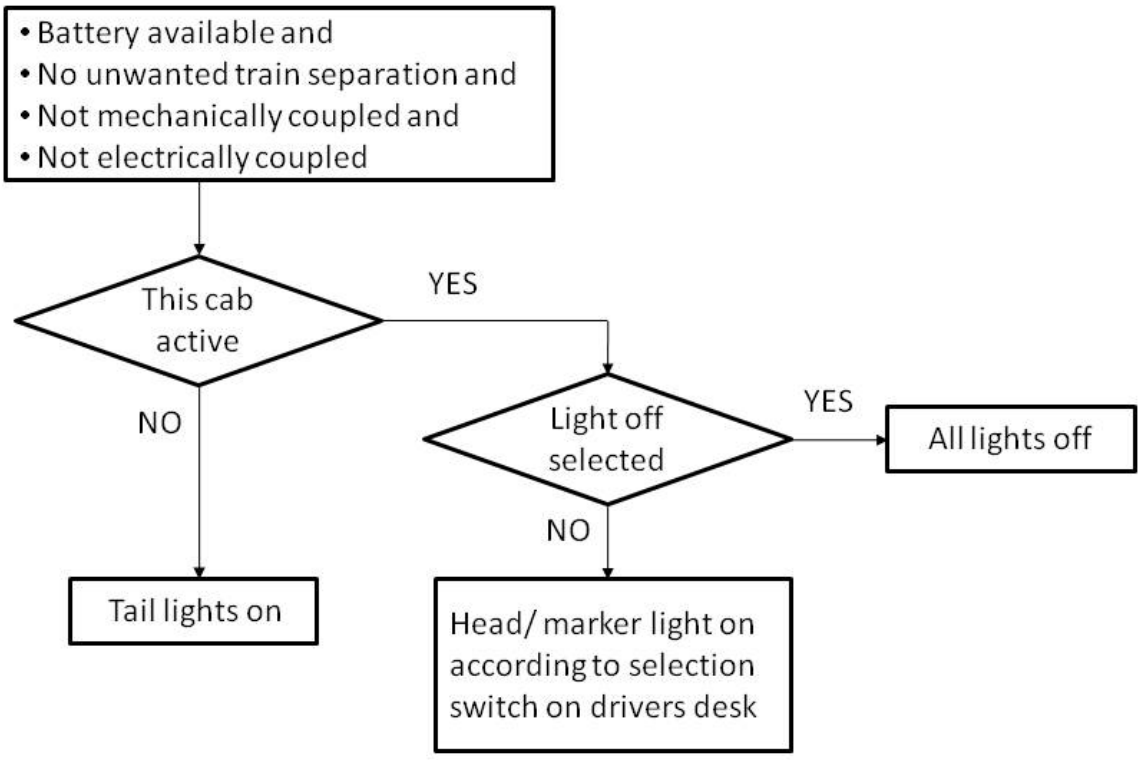
3 Functional requirements

3.1 Function Front and rear lamps control

3.1.1 Function Design

In general tail lamps on both ends of the train are active whenever battery power is available. The head and marker lights are activated by a switch positioned on the driver's desk. The light shall not be active at an intermediate point of the train(multi traction). In addition, the light must not be switched on at intermediate point if the train is unwanted separated. The light attached to the active cab shall be white and the light at the tail of the complete train shall be red.

The design described in this SFRS refers only to one light system at one cab and covers marker, tail and head lights. The head and marker light is controlled by a switch on the driver's desk of the attached cab. The tail light is controlled automatically.



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3.1.1.1 Normal condition

The trains set shall be marked in all standing modes apart from driving mode with selection switches in the position "off" with tail lights on at the front and the end whenever battery power is available. When the train set is in driving mode, the marker and headlights at the front of the train shall act according to the selected light case.

3.1.1.2 Failure Case

1. Power supply at fault
2. Single light unit degraded - loss of single branch of LED
3. Single light unit at fault - loss of more than one branch of LED
4. Multiple light units of same type degraded - loss of single branch of LED in more than one marker light unit
5. Multiple light units of same type at fault - loss of more than one branch of LED in more than one marker light unit

3.1.1.3 Failure Detection

1. Power supply signal is not set when light function is requested
2. Diagnostic for effected group of light function set.
3. No detailed detection possible during service.
4. No detailed detection possible during service.
5. No detailed detection possible during service.

3.1.1.4 Failure Handling

1. Driver to check other adequate light functions.
2. Set of normal diagnostic event. Replacement of light unit during scheduled workshop visit
3. Set of normal diagnostic event. Replacement of light unit during scheduled workshop visit
4. Set of normal diagnostic event. Determination of effected light unit during workshop visit. Replacement of light unit during scheduled workshop visit.
5. Set of normal diagnostic event. Determination of effected light unit during workshop visit. Replacement of light unit during scheduled workshop visit.

3.1.1.5 Failure Effect

1. If adequate light is available, service is possible. If not, service is not allowed
2. No effect on service
3. No effect on service
4. No effect on service
5. No effect on service

3.1.2 Requirements

ID	2F_02.06.Zefiro-Europe.TRS.12
Description	The driver shall be able to control the head and marker lamps from the normal driving position.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.113	A control device with 3 stable positions shall be installed on the drivers desk to control the side head lights with the functions A) side head light - off B) side head light - full beam selected C) side head light - dimmed selected	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.06.Zefiro-Europe.CONCEPT.140	The side head light control device shall be exclusively operable when the cab is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A

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2F_02.06.Zefiro-Europe.CONCEPT.76	A control device with 3 stable positions shall be installed on the drivers desk to control the marker lights with the functions D) marker lights white - off E) marker lights white - full selected F) marker lights white - dimmed selected	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.139	The marker light control device shall be exclusively operable when the cab is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.115	A control device with 2 stable positions shall be installed on the drivers desk to control the head light function of the centre light G) centre head light - off H) centre head light - selected	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.06.Zefiro-Europe.CONCEPT.142	The control device of the centre head light shall be exclusively operable when the cab is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_02.06.Zefiro-Europe.CONCEPT.146	A control device for the function "running on prohibited rail" shall be included on the drivers desk. I) function selection - prohibited rail	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.06.Zefiro-Europe.CONCEPT.158	The control device for "running on prohibited rail" shall be exclusively operable when the cab is active and overrule every selection from the marker light control device.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.108	The selection of the exterior lighting control devices A) side head light - off B) side head light - full beam selected C) side head light - dimmed selected D) marker light white - off E) marker lights white - full selected F) marker lights white - dimmed selected G) centre head light - off H) centre head light - selected I) function selection - prohibited rail shall be submitted to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.123	The ext. lighting system shall have 7 inputs 1) head light - full beam selected 2) head light - dimmed selected 3) marker lights white - full selected 4) marker lights white - dimmed selected 5) marker lights red - selected 6) centre head light - selected 7) driving on prohibited rail	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A
2F_02.06.Zefiro-Europe.CONCEPT.164	The ext. lighting system shall have 7 inputs 1) head light - full beam selected 2) head light - dimmed selected 3) marker lights white - full selected 4) marker lights white - dimmed selected 5) marker lights red - selected 6) centre head light - selected 7) driving on prohibited rail	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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2F_02.06.Zefiro-Europe.CONCEPT.133	The ext. lighting system shall have 4 diagnostic outputs 1) side head light at fault 2) marker light white at fault 3) marker light red at fault 4) centre head light at fault	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A
2F_02.06.Zefiro-Europe.CONCEPT.165	The ext. lighting system shall have 4 diagnostic outputs 1) side head light at fault 2) marker light white at fault 3) marker light red at fault 4) centre head light at fault	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.137	The ext. lighting system shall have 1 status output 5) power supply on/off	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A
2F_02.06.Zefiro-Europe.CONCEPT.166	The ext. lighting system shall have 1 status output 5) power supply on/off	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.135	The ext. lighting system out puts shall only be set if the corresponding light function is requested	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.143	The ext. lighting system status output shall be sent to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.136	If "marker light white dimmed" (OTC_BEAM_HL2/4) is selected and "marker light white at fault" (OTC_BEAM_ML_WH) is set, an EVENT shall be set stating "white marker light dimmed is at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.172	If "marker light white full" (OTC_BEAM_HL4/4) is selected and "marker light white at fault" (OTC_BEAM_ML_WH) is set, an EVENT shall be set stating "white marker light full is at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.173	If "centre head light" (OTC_BEAM_HLCENT) is selected and "centre head light at fault" (OTC_BEAM_CHL) is set, an Event shall be set stating "centre head light is at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.174	If "head light dimmed" (OTC_FBEAM_HLDIM) is selected and "side head light at fault" (OTC_BEAM_HL) is set, an EVENT shall be set stating "Dimmed head light is at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.175	If "head light full beam" (OTC_FBEAM_HLFLASH) is selected and "side head light at fault" (OTC_BEAM_HL) is set, an EVENT shall be set stating "full beam head light is at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.176	If this cab is not active, the cab is not coupled and the "marker light red at fault" (OTC_BEAM_ML_RD) is set, an EVENT shall be set stating "red marker light at fault"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.177	If the normal control of the light system is blocked by the rotary switch, all diagnostic events shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_02.06.Zefiro-Europe.CONCEPT.152	If a light function is selected and the power supply status is off, an event shall be set by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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ID	2F_02.06.Zefiro-Europe.TRS.14
Description	The driver shall be able to turn all lamps off
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.74	If function A of the ext. lighting system is selected, the ext. lighting system inputs 1 and 2 shall not be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.110	If function D of the ext. lighting system is selected, the ext. lighting system inputs 3 and 4 shall not be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.159	If function G of the ext. lighting system is selected, the ext. lighting system input 6 shall not be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_02.06.Zefiro-Europe.TRS.11
Description	The driver shall be able to turn dimmed marker lamps on
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.75	If function F is selected, the ext. lighting input 4 shall be set and the input 3 shall not be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.91	Marker lights and centre light shall have a dimmed and a turned up light mode	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A
2F_02.06.Zefiro-Europe.CONCEPT.112	centre light shall have a dimmer and turned up marker light mode and a dimmed head light mode	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A

ID	2F_02.06.Zefiro-Europe.TRS.19
Description	The driver shall be able to turn full marker lamps on
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.83	If function E is selected, the ext. lighting input 3 shall be set and the input 4 shall not be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

ID	2F_02.06.Zefiro-Europe.TRS.16
Description	The driver shall be able to turn dimmed headlamps on
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.84	If function C is selected, the ext. lighting input 2 shall be set and the input 1 shall not be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.92	Head lights shall have a dimmed and a full beam mode	Derived Requirement	4S_08.05.02.-.Exterior Lights Systems		N/A

ID	2F_02.06.Zefiro-Europe.TRS.22
Description	The driver shall be able to turn full-beam headlamps on
Safety level	N/A

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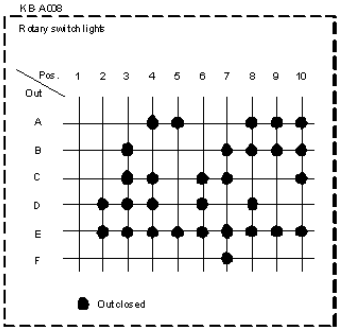
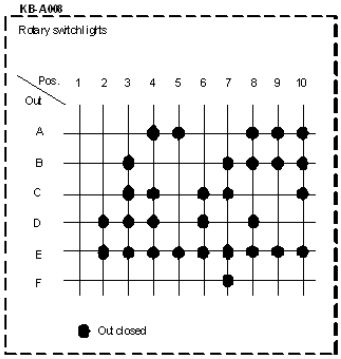
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.85	If function B is selected, the ext. lighting input 1 shall be set and the input 2 shall not be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

The centre light unit shall have the functionality of a dimmed head light

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.161	If function H is selected, the ext. lighting input 6 shall be set	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_02.06.Zefiro-Europe.TRS.23
Description	In addition to their traditional role as front and rear lamps, the lamps may be used in specific ways and arrangements in emergency.
Safety level	N/A

The ext. lighting system shall have a function for running on prohibited rail as requested by the RFI

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.163	A 10-position rotary switch shall be integrared on the drivers desk and linked to TCMS via 4 digital inputs (rotary switch position indication is described by SCM002761) 	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.178	The rotary switch shall be linked to TCMS via the following digital inputs: A OTC_BEAM_SWITCH1 B OTC_BEAM_SWITCH2 C OTC_BEAM_SWITCH3 D OTC_BEAM_SWITCH4 	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_02.06.Zefiro-Europe.CONCEPT.179	Rotary switch will be indipendent from cab active. The position of rotary switch is taken in account only if the cab is not on a coupled car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.06.Zefiro-Europe.CONCEPT.180	When the driver's desk is activated TCMS shall check the position of the rotary switches in all of the not activated cabs. If any of the rotary switches in the not activated cabs is not in position 1, a warning will be shown to the driver and the driver shall acknowledge it.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_02.06.Zefiro-Europe.CONCEPT.169	If position 1 is selected, the control of the light units shall be done via the the switches on the drivers desk. In any other position, the control via the switches on the drivers desk shall be blocked.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_02.06.Zefiro-Europe.CONCEPT.171	The block of the "normal" control shall be linked to a digital input to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.170	The block of the "normal" light control function shall be supervised by TCMS (OTC_Light_RSwitch)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.06.Zefiro-Europe.CONCEPT.167	If the switch is set in position 2 TCMS shall activate the "Running on prohibited rail" function (TCMS_BEAM_COM5)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_02.06.Zefiro-Europe.CONCEPT.168	If position 3 is selected, TCMS shall activate the digital outputs for the dimmed head lights (TCMS_BEAM_COM1 and TCMS_BEAM_COM2)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_02.06.Zefiro-Europe.TRS.33
Description	The tail lamps shall be automatically switched on when: the dimmed marker lamps are turned on or the full marker lamps are turned on or the dimmed headlamps are turned on or the full-beam headlamps are turned on. This requirement shall not be applicable for train ends that are coupled.
Safety level	SIL 0

ID	2F_02.06.Zefiro-Europe.TRS.28
Description	The external lamps located at intermediate points within the train shall be switched off.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.86	If Battery is available and the cab is not active the ext. lighting input 5 shall be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.77	If the train is mechanically coupled or electrically coupled at the attached coupler, no light shall be active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.90	If the train is unwanted separated at the attached coupler, no light shall be active	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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ID	2F_02.06.Zefiro-Europe.TRS.52
Description	The use of lights to inform of an emergency situation (operating rule, see TSI OPE) should be done only by means of head lamps in flashing/blinking mode.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_02.06.Zefiro-Europe.CONCEPT.182	A lighted bi-stable control device shall be installed on the drivers desk to control the head light (full beam) blinking function	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_02.06.Zefiro-Europe.CONCEPT.187	The lighted bi-stable control device shall be tested whent test lamp is active	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.183	The postion bi-stable control device shall be notified to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.184	If the bi-stable control device is activated then an 'other' event shall be set to indicate that the blinking is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_02.06.Zefiro-Europe.CONCEPT.185	The blinking fuction shal be implemented with 120 cycles/min, 1/3 on 2/3 off	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.186	The blinking fuction shall be active only in France	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.189	The diagnostic output of the UNYSYS control box (CN1-f) shall be disconnected when the blinking function is activated	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_02.06.Zefiro-Europe.CONCEPT.188	When blinking function is active the head lights command (full and dimmed) shall be cut-out	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.28	side head light - full beam selected	L	2F_02.06.Zefiro-Europe.CONCEPT.74 2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.84 2F_02.06.Zefiro-Europe.CONCEPT.85 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.140 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.124	side head light - dimmed selected	L	2F_02.06.Zefiro-Europe.CONCEPT.74 2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.84 2F_02.06.Zefiro-Europe.CONCEPT.85 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.140 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.125	marker lights white - full selected	L	2F_02.06.Zefiro-Europe.CONCEPT.75 2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.83 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.110 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.139 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.126	marker lights white - dimmed selected	L	2F_02.06.Zefiro-Europe.CONCEPT.75 2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.83 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.110 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.139 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.127	marker lights red - selected	L	2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.86 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.128	centre head light - selected	L	2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.142 2F_02.06.Zefiro-Europe.CONCEPT.159 2F_02.06.Zefiro-Europe.CONCEPT.161 2F_02.06.Zefiro-Europe.CONCEPT.164
2F_02.06.Zefiro-Europe.CONCEPT.151	function: prohibited rail	L	2F_02.06.Zefiro-Europe.CONCEPT.77 2F_02.06.Zefiro-Europe.CONCEPT.90 2F_02.06.Zefiro-Europe.CONCEPT.123 2F_02.06.Zefiro-Europe.CONCEPT.158 2F_02.06.Zefiro-Europe.CONCEPT.163 2F_02.06.Zefiro-Europe.CONCEPT.164

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Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.29	white marker light at fault	L	2F_02.06.Zefiro-Europe.CONCEPT.133 2F_02.06.Zefiro-Europe.CONCEPT.165
2F_02.06.Zefiro-Europe.CONCEPT.129	side head light at fault	L	2F_02.06.Zefiro-Europe.CONCEPT.133 2F_02.06.Zefiro-Europe.CONCEPT.165
2F_02.06.Zefiro-Europe.CONCEPT.130	red marker light at fault	L	2F_02.06.Zefiro-Europe.CONCEPT.133 2F_02.06.Zefiro-Europe.CONCEPT.165
2F_02.06.Zefiro-Europe.CONCEPT.131	centre head light at fault	L	2F_02.06.Zefiro-Europe.CONCEPT.133 2F_02.06.Zefiro-Europe.CONCEPT.165
2F_02.06.Zefiro-Europe.CONCEPT.132	power supply on/off	L	2F_02.06.Zefiro-Europe.CONCEPT.137 2F_02.06.Zefiro-Europe.CONCEPT.166

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.116	dimmed marker light selected	L	2F_02.06.Zefiro-Europe.CONCEPT.75 2F_02.06.Zefiro-Europe.CONCEPT.76 2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.152
2F_02.06.Zefiro-Europe.CONCEPT.117	full marker light selected	L	2F_02.06.Zefiro-Europe.CONCEPT.76 2F_02.06.Zefiro-Europe.CONCEPT.83 2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.152
2F_02.06.Zefiro-Europe.CONCEPT.118	dimmed side head light selected	L	2F_02.06.Zefiro-Europe.CONCEPT.84 2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.113 2F_02.06.Zefiro-Europe.CONCEPT.152
2F_02.06.Zefiro-Europe.CONCEPT.119	full beam side head light selected	L	2F_02.06.Zefiro-Europe.CONCEPT.85 2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.113 2F_02.06.Zefiro-Europe.CONCEPT.152
2F_02.06.Zefiro-Europe.CONCEPT.120	dimmed centre head light selected	L	2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.115 2F_02.06.Zefiro-Europe.CONCEPT.152 2F_02.06.Zefiro-Europe.CONCEPT.161
2F_02.06.Zefiro-Europe.CONCEPT.150	function: prohibited rail - selected	L	2F_02.06.Zefiro-Europe.CONCEPT.108 2F_02.06.Zefiro-Europe.CONCEPT.146 2F_02.06.Zefiro-Europe.CONCEPT.152
2F_02.06.Zefiro-Europe.CONCEPT.153	Side head light at fault		2F_02.06.Zefiro-Europe.CONCEPT.136 2F_02.06.Zefiro-Europe.CONCEPT.143
2F_02.06.Zefiro-Europe.CONCEPT.154	Tail light at fault		2F_02.06.Zefiro-Europe.CONCEPT.136 2F_02.06.Zefiro-Europe.CONCEPT.143
2F_02.06.Zefiro-Europe.CONCEPT.155	Centre head light at fault		2F_02.06.Zefiro-Europe.CONCEPT.136 2F_02.06.Zefiro-Europe.CONCEPT.143
2F_02.06.Zefiro-Europe.CONCEPT.156	white marker light at fault		2F_02.06.Zefiro-Europe.CONCEPT.136 2F_02.06.Zefiro-Europe.CONCEPT.143
2F_02.06.Zefiro-Europe.CONCEPT.157	power supply on/off		2F_02.06.Zefiro-Europe.CONCEPT.143 2F_02.06.Zefiro-Europe.CONCEPT.152

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Outputs:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_02.06.Zefiro-Europe.CONCEPT.33			

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